Structures and Architecture. REstructure REmaterialize REthink REuse - Rinke & Frier Hvejsel (Eds)
© 2025 The Author(s), ISBN 978-1-041-11139-9
Open Access: www.taylorfrancis.com, CC BY-NC-ND 4.0 license

Balancing resources and cultural values in building adaptations

M. Reffs Kramhøft, P. Munch-Petersen & H. Ejstrup
The Royal Danish Academy, Copenhagen, Denmark

ABSTRACT: This article examines the relationship between cultural heritage and environmental impact in building adaptations through a proposed matrix combining SAVE (Survey of Architectural Values in the Environment) and LCA (Life Cycle Assessment) methodologies. It indicates that a direct methodological merge of the systems increases the awareness of both ecological and cultural values and sharpens priorities and approaches when adapting existing structures for new use. Initial case studies suggest that buildings with high preservation values tend to achieve lower environmental impacts when adapted, while buildings with low preservation values, despite offering greater adaptation potential, often lead to more extensive interventions with higher environmental impacts. The study reveals a significant correlation between the systems, suggesting that traditional building approaches and restrained intervention strategies may offer valuable insights for more thoughtful and ethical solutions. The proposed matrix serves as an analytical benchmarking tool for integrating cultural and ecological values in early-stage architectural design decisions.

1 INTRODUCTION

Preserving, reusing, and transforming existing buildings are key in the green transition of construction, as they represent significant resources and embedded CO2 (IPCC, 2023) Yet, existing buildings refer to many value systems besides the ecological ones. Buildings also represent cultural values and architectural heritage, which involve historical aspects that infuse them with meaning, character, and a sense of belonging in a place. Architectural styles and an archaeological approach have traditionally guided our view of buildings worthy of preservation (Venice Charter, 1967) However, the growing emphasis on preserving structures has led to increasingly diverse approaches to building adaptations. Modern interventions must balance the purpose of preserving building heritage while also reusing resources and adapting buildings to meet society's changing needs (Plevoets & Van Cleempoel, 2019). A new Danish study further shows significantly and unambiguously that renovation is better for the climate than demolition and new construction (Realdania, 2024).

In Denmark, Life Cycle Assessment (LCA) by legislation has been introduced as a standard method for estimating CO2 emissions ((Danish Building Regulations, 2024). LCA presents a fundamental but narrow view of the environmental impacts since it does not include all planetary boundaries and has a somewhat arbitrary scope of 50 years with no regard to the type, placement, or ambition level of the architecture (Munch-Petersen, 2020). Regarding architectural heritage, assessment systems like the Danish method; Survey of Architectural Values in the Environment (SAVE) focus on buildings' cultural and historical aspects (Didriksen, Tønnesen, Høi & Stenark, 2011), but they pay no attention to the climate agenda. Neither refers to the social characters of place or community level nor do they touch upon phenomenological aspects relating to buildings.

It can be argued that the value systems inevitably shape how we understand and prioritize intervention processes. This raises important questions: How can these different value paradigms

DOI: 10.1201/9781003658641-51

coexist in building adaptations? Do they inherently conflict or can they be strategically aligned to reinforce each other? To investigate these questions and test the balance between the value systems a matrix is set up merging SAVE and LCA. The purpose of the tool is to bring attention to both methodologies informing preliminary phases and being able to make informed decisions before and in the design process.

In this paper the term ***adaptation*** is used to describe an extensive change to a building and *"any work to a building that goes over and beyond maintenance to change its capacity, function, or performance"* (Douglas, 2006). The word ***intervention*** is used as a more neutral but concrete act or process of intervening in and working with any type of existing structure in general.

2 METHOD

2.1 *Assessment methods and building adaptation strategies*

Assessments of existing buildings and their context are crucial when intervening in them. Evaluating inherent values based on different systems, along with more objective technical registrations and preparatory investigations, establishes the foundational core values. With new needs and functions, this forms the basis for selecting the appropriate approach and design strategies when adapting old buildings to new use (Cramer & Breitling, 2007). Interventions in existing buildings is a field of many professions and research areas and it presents a complex challenge that intersects multiple disciplines, demanding a nuanced understanding of historical, environmental, and technical considerations (Plevoets & Van Cleempoel, 2019). The common purpose is the process of working with an existing situation and the ambition of retaining (some of) the original structure, resources, and collective memories related to the place (Stone, 2019). The process demands a different way of thinking and seeking what inherent possibilities the existing building has to offer rather than dictating what it has to be (Cramer & Breitling, 2007).

But the desire to preserve can be ambivalent. On one hand, we strive to conserve specific historic structures, and on the other, the aging process is unstoppable. *"Because their structure tends to outlive their function, buildings have continuously been adapted to new uses – a fact which has enabled generation after generation to derive a sense of continuity and stability from their physical surroundings."* (Cantacuzino, 1975). This observation underscores the historical practice of building reuse, driven by both pragmatic and symbolic considerations. And that preservation is no contradiction to change in the built environment. Adaptations of existing buildings are not a static or retrospective approaches, as attitudes develop as well. Even with the same facts and details, our interpretation and perspectives on the subject may well adjust (Strike, 1994).

In this ephemeral reality where attitudes change over time, common frameworks guide us to point out values for the further process. The tools and methodologies we employ in assessing and adapting existing buildings significantly influence our approach to preservation, serving as crucial guides in decision-making processes (Stylsvig Madsen, Beim & Reitz, 2015).

2.2 *Life Cycle Assessment – A new value system*

On the 5th of March 2021, the Danish parliament voted for an agreement to implement the National Strategy for Sustainable Construction (Ministry for Interior & Housing, 2021). It was initially implemented in January 2023 as an expansion to the Danish Building Code (Danish Building Regulations, 2018). This means that from 2023 onward all new construction must have a life cycle assessment {LCA} as part of the building permit application. It was implemented with benchmarks for different sizes of buildings. As part of the agreement, these benchmarks should be adjusted continuously, ensuring an increasingly ambitious environmental aim for new construction over time.

On the 20th of May 2024 a new agreement defined new benchmarks for construction with values per typology this time around. Dwelling, offices, institutions, and more have different specific benchmarks that are heightened in 2025, 2027, and 2029 (Ministry of Social Affairs & Housing, 2024).

Building regulations have long incorporated various performance requirements that have become ingrained in standard construction practice to such a degree that we hardly think of them as value systems (Munch-Petersen & Ejstrup, 2017). Chief among these is the steadily increasing energy efficiency mandate, originally driven by the 1970s oil crisis and political aims to reduce OPEC dependency. Today, these standards align with the Paris Agreement and broader environmental sustainability goals. A building's value is now intrinsically tied to its calculated energy performance - if it fails to meet the standards, construction approval will be denied (Danish Building Regulations, 2018, chapter. 11) Energy efficiency has effectively become a core value system for contemporary construction.

The climate crisis has now introduced another value system in the form of LCA. It focuses primarily on Global Warming Potential (GWP), measured in CO2-equivalents, as the primary environmental indicator based on the Paris Agreement's priorities (United Nations, 2015). Other common LCA metrics like Ozone Depletion, Photochemical Smog, Acidification, and Eutrophication are not specifically targeted. Sustainable construction is thus defined through benchmarked, relativistic LCA measures addressing only GWP.

The mandatory inclusion of LCA in the building code signals a new mindset - sustainability is now a compulsory value system for all new construction. As the LCA benchmarks become more stringent over time, a sense of urgency is implied. However, this sustainability framework does not currently extend to existing buildings, where conservation, renovation, and adaptation projects remain exempt from LCA requirements. This represents a significant gap in harmonizing cultural and environmental values across the built environment (Schjødt Worm, 2016).

2.2.1 *A (new) focus on the materials of architecture*

LCA is done on material/component level. As such, the value system LCA represents is a material value system. Some material represents high GWP and some less (Munch-Petersen & Ejstrup, 2017). The 'right' material of architecture then becomes a question of environmental loads of materiality and not functionality, cost, aesthetics, and to a lesser degree performance – and this represents a change in view on architectural material performance.

As a value system, LCA dictates a new focus on low-impact materials, and this has a significant influence on how architecture is understood and performed. In public architectural discourse the idea of 'new-construction-stop' and 'demolition-stop' is born from this new perspective. Adaptive reuse and material reuse are ways to make sure that the materials in use in construction stay in use. From an LCA perspective, this means that reused materials and architectural compositions represent zero impact according to the specifications of LCA calculation. Finding aesthetic value in construction that is already there has become key and is becoming a mainstream idea in sustainable thinking in construction.

2.3 *SAVE – Architectural historic value*

Survey of Architectural Values in the Environment (SAVE) is a method to assess architectural, cultural-historical, and environmental (not ecological) values in buildings, lands- and townscapes to identify buildings and cultural environments worthy of preservation (Didriksen, Tønnesen, Høi & Stenark, 2011). The system is focused on 'the bigger picture' rather than on the individual building, as it can analyze a large number of buildings and settlements within a short time (Stylsvig Madsen, Beim & Reitz, 2015).

SAVE emerged from the evolution of preservation strategies both in Denmark and internationally. Denmark's Building Preservation Act (Bygningsfredningsloven) of 1918 initially established two listing categories: A-listings (interior and exterior) and B-listings (exterior only), which were merged in 1980. The post-World War II period saw increased interest in heritage preservation, marked by the establishment of numerous preservation associations (Bendsen, Morgen & Lindhe, 2018). Denmark's joining of the Granada convention in 1985 was significant, as it promoted architectural heritage preservation across Europe (Council of Europe, 1985). While the Building Preservation Act could address buildings and their immediate surroundings, it couldn't protect independent landscape elements like parks and cityscapes.

In response, the Ministry of Environment and Energy and The National Forest and Nature Agency instigated the development of SAVE in 1987. Launched in 1990-92 as an open-source method, it led to the creation of 90 municipal atlases documenting local heritage and cultural identity (Didriksen, Tønnesen, Høi & Stenark, 2011). The methodology went international with the 1997 publication of InterSAVE (Didriksen & Tønnesen, 1997). Today, buildings deemed worthy of preservation through SAVE are regulated by municipalities under The Act of Planning (Planloven, 2018).

2.3.1 *Methodology and legal framework*

SAVE is built upon 5 criteria for assessment: Architectural value, environmental value, cultural-historical value, originality, and technical condition. The final SAVE value is a weighted summarization of the 5 criteria (Didriksen, Tønnesen, Høi & Stenark, 2011). SAVE has a twofold focus: 1) Environmental structures (built or landscape) and 2) Single Buildings. When SAVE is used for assessing environmental structures, a more rigorous process for registration is being commenced, than when assessing single buildings, but the 5 criteria are the same (Didriksen, Tønnesen, Høi & Stenark, 2011). Furthermore, the methodology of the 5 criteria is also reflected in the Assessment of Conservation Values used when assessing and describing the listed buildings hereby creating a coherent assessment methodology across both listed buildings and buildings worthy of preservation (Agency of Castle and Cultures, 2010).

When SAVE is used for assessing buildings, the screening is very quick, 10-20 minutes including preparations/transport, description, photography of the facades, and reporting. SAVE only focuses on the present state of the building on a local/regional scale and therefore not comparable across the different regions (Didriksen, Tønnesen, Høi & Stenark, 2011). The assessment is not legally binding in itself unless the building is recorded in a local plan or municipal plan as a building worthy of preservation (Bygningsfredningsloven, 1986). In this case, the owners are legally obliged to obtain a permit from the municipality, before changes or building works can be commenced on the facade of the building. Furthermore, the municipality can ask for further specifications, and documentation or impose constructional or material conditions before a permit can be obtained. Also, a building worthy of preservation cannot be demolished without the case being accepted by the city council based on a public hearing and illegal demolitions can result in legal action and compensation claims.

3 AN ECOLOGICAL-HERITAGE VALUE MATRIX

3.1 *Combining the two value systems*

It is clear that SAVE and LCA are different value systems but both are important systems to gauge the cultural significance and the ecological impacts of building now and over time. The question then is whether the climate agenda or the architectural heritage agenda can coexist in one assessment tool. If they have a common language and they can be understood as two sides of the same value assessment? To explore this discrepancy or coherence of the two systems within building adaptation, a simple matrix system has been developed (see Figure 1). As both systems deal with a quantitative methodology a mathematical matrix can be an obvious choice as a point of departure for this inquiry. It combines *culture-scores* with *ecology-scores* in a system where it's not an either/or scale, but an emphasis and balance between the two. The matrix operates with quantifiable values representing the number span for the two systems. By simply showing both results simultaneously the idea is that both value systems can be addressed at the same time.

Both LCA and SAVE are valuation tools that use simplified approaches to complex relations and interdependencies of construction. SAVE is based on both qualitative descriptions and a final grading, which indicates the overall preservation value. Similar to an LCA assessment, the final score in both systems does not cover the comprehensive assessment but is an attempt to provide a single simplified overall statement. In other words, something we might best describe qualitatively becomes something that can be given a score, which, in turn, can be useful for comparisons. Therefore, knowing that none of the value systems manage to embrace the complexity of

architecture fully, they are today embedded in the Danish systems and legal framework and the only valid practice-based assessment tools we can employ.

The respective scales in the matrix are the score 1-9 aligned with SAVE and the LCA score span between 0 to 12 kg CO2-eq/m2/year on the vertical line, which refers to the threshold limit value per 2024 (Ministry for Interior & Housing, 2021). The results from LCA and SAVE assessments are placed in relation to each axis in the matrix. Structuring the diagram this way allows for the two systems to complement each other rather than being opposites or done independently. By combining the two systems the discussion about the architectural heritage also becomes an ecological discussion on how much we can afford to preserve or demolish when the ecological budget must not be exceeded. This could foster a relationship where the two systems strengthen one another in practice.

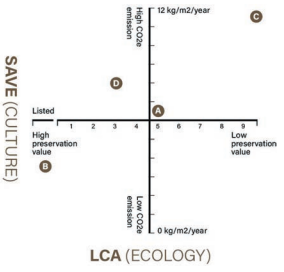


Figure 1. Value matrix including placement of cases.

3.2 *Testing the matrix*

It is clear that the two value systems have contrasting temporal perspectives and come from two different time periods - one focusing back on the value of existing architectural conditions and the other on environmental impacts now and in the future. This is especially evident when it comes to data availability. Only a few buildings have undergone both assessments, largely because LCA is not yet a legal requirement for adaptive reuse and renovation projects (Schjødt Worm, 2017). But also, because systematic assessment of post-1940s buildings only began recently, leading to limited database entries for more recent architecture (Realdania, 2024). To connect those contrasting foci, using the matrix at multiple stages throughout the design process could prove valuable: initially to understand the inherent value before intervention and set ambition levels; then to evaluate how different strategies might affect both indicators; and finally, to assess whether the project succeeded in preserving architectural value while minimizing environmental impact. As an initial test, four cases were found and placed in the matrix (see Figure 1). Three of them have a SAVE and a LCA made in accordance with common practice for SAVE and LCA in Denmark. The last one (case C) does not have a SAVE but is treated in its transformation as being of low preservation value. As such all four cases are methodologically compatible. In the following, the cases are described and placed in the matrix.

A) ***A functionalistic office building*** from 1957, built in classic modernistic prefab concrete elements and situated in the heart of Copenhagen. It holds a formal SAVE value of **5**, characterized as



Figure 2. Case A-B-C-D, Credits: A+C *Henning Larsen*, B *Agency for Culture and Palaces,* D *Peter Elfelt.*

a middle preservation value. The actual assessment is from 1995 and with very limited information. In 2021 the roughly 12.000 m2 building was reused through a deep renovation to create a modern, energy-optimized mixed-use building adapted to contemporary standards. The original concrete facade was replaced with a new sandstone facade and the interior of the building was in the rebuilding process stripped down to the raw structure. The result for the final in the LCA assessment was **6,5** kg CO2e/m2/year.

B) ***A historic farmhouse*** dating back to different periods between 1778-1853 and a part of a preservation-worthy and cohesive village environment. A traditional four-lane farm with a thatched roof and timber frame construction, and it contains constructional detailing and materials used for the specific region. The building itself is **listed** and therefore subject to special restrictions. For many years the farm had been neglected and the condition was such that the deterioration was accelerating and without intervention, there was a risk of collapse. By using solely reused materials, new biobased ones, and change to a geothermal heating source, the intervention managed to gain a final LCA result of **3,3** kg CO2e/m2/year.

C) ***A former court and police station*** is transformed into a new health and care centre. The 4.300 m2 building from 1973 is a characteristic rational prefabricated concrete element construction with a great deal of repetition. It does not hold a SAVE value itself but is placed in an area designated as worthy of preservation. The building has been stripped down to its structure. The ambition for the client is to do an extension to the existing building and establish a coherent whole in a completely new contemporary expression hiding the original building. The project is assessed to have an LCA value of **11,2** kg CO2e/m2/year.

D) ***A classical Copenhagen block*** from 1893, placed in one of the city's most central and prominent locations. Its approximately 7.000 square meters have contained changing residential and commercial functions throughout its lifetime. The building is constructed with a base of sand-lime blocks and top of red bricks and with extensive use of ornamentation that was characteristic for the historicistic period. The SAVE value assessment was done in 1995 with a score of 3, but holds no further description. Besides both conservation and renovation of the existing parts, new construction is added up- and inwards adapting the building to fit new purposes. The project is estimated to reach a LCA value of **8,1** kg CO2e/m2/year.

4 DISCUSSION

The matrix is primarily designed to inform early-phase decisions, focusing on initial intervention ideas while incorporating LCA considerations, which require detailed material knowledge. This creates a process where cultural preservation values must align with LCA score targets and vice versa. Additionally, the matrix serves as a benchmarking tool to evaluate different design approaches, revealing their ecological 'costs' and helping identify preferable strategies regarding environmental impact. As such the studied cases are not optimal since SAVE assessments were completed before design proposals, while LCA was conducted post-intervention. As a design tool, the matrix should be tested multiple times during early design stages, ensuring teams maintain awareness of both cultural heritage and ecological concerns throughout the process.

Higher SAVE values impose greater legal constraints and limit building adaptation freedom. This protects heritage authenticity from radical changes, following the principle that less intervention means less damage. It can also reduce environmental impact by minimizing resource use, as demonstrated in ***Case B***, where a high SAVE score correlates with a relatively low LCA score. The preservation requirements limit unnecessary interventions, and the use of biobased materials compatible with the original construction helps maintain ecological boundaries. Here the initial ambition of ending up with a low LCA score made the design team choose low-impact solutions for the adaptations - showcasing both value systems at play in the design process.

Conversely, buildings with lower heritage value offer more freedom to intervene. The adaptation potential increases as architectural heritage value decreases. With fewer constraints and emphasis on use and resource value, there are fewer limits on structural modifications and material changes, as seen in ***Case C*** where the low value opens up all possible approaches and design

strategies. However, if the matrix had been used early in design, the LCA score might have been lower as ecology would have been a visible design parameter.

Case A demonstrates how focusing on operational energy carbon savings helped meet the standards of new builds while the preservation value was important for keeping the original facade expression. ***Case D*** significantly shows how preservation and adaptation can coexist, with street-facing facades carefully conserved due to a high SAVE score, while allowing roof extensions and courtyard modifications to create new relevance for the building.

The examples demonstrate that greater interventions typically create larger carbon footprints, aligning with literature on renovation and building adaptations (Realdania, 2024). It indicates potential CO2 savings of up to 40% through reuse rather than recycling (Andersen, 2023), suggesting that maintaining materials in existing buildings is the most effective resource preservation strategy. The initial tests also indicate that a high SAVE score instills a will to do as little as possible in the design team. Suggesting that high cultural value, in itself, leads to more ecological thinking. This needs to be tested further in the coming research.

5 CONCLUSION

This article has aimed to examine the relationship between cultural and ecological value systems and how their integration could inform and improve our approach to building adaptations. While the proposed matrix uses methods that offer limited views of their respective fields, understanding the balance between ecology and culture in building assessment could foster more dependent cohesion between these value systems.

Although the matrix has not yet been implemented as an early-stage design tool, the initial tests in this article suggest that cultural and ecological values can successfully coexist in a quantitative relation. The matrix serves as an effective benchmarking tool, allowing comparisons between projects with different ambitions.

The cases in the study show that adaptation of buildings with a high preservation value (SAVE score) does achieve lower environmental impacts (LCA scores). It also shows that buildings with low preservation value may offer rebuild potentials to a degree where missing legal limitations could lead to higher environmental impacts. As more data come to light it will be interesting to see if a larger empirical dataset could verify this result and how further ecological priorities might influence future adaptation projects.

Interestingly, our search for solutions to current ecological challenges often leads us back to architectural heritage which points to a significant correlation between the two systems. Pre-industrial, biobased building approaches may offer insights into contemporary sustainable design as showcased in case B. At the same time, more humble and thoughtful approaches to all buildings - where as much material as possible remains and where the existing fabrics shows the future possibilities - may prove to be a good strategy for both minimizing environmental impacts and respecting the architectural heritage. Understanding that buildings emerge from specific contexts and changing socio-cultural conditions allows us to take a more comprehensive approach to their entire lifecycle. This includes appreciating their values and understanding their impacts while adapting them to evolving societal needs. The matrix attempts to visualize the value systems simultaneously, helping integrate the understanding into early architectural design phases.

6 PERSPECTIVES

The preliminary testing of juxtaposing SAVE and LCA assessment systems highlights the need for more extensive evaluation across a broader range of adaptation projects to determine the generalizability of the findings.

The use of the SAVE methodology in the matrix presents several limitations. Beyond its single-score approach that excludes qualitative descriptions, the system was not originally designed to holistically capture cultural aspects. It's simply not what SAVE is developed to do. Furthermore,

outdated SAVE assessments may fail to reflect contemporary attitudes and neglect important aspects of the self-grown that have been formed over time representing identity and narratives. While SAVE is effective for screening historic buildings, it is less suitable for evaluating final adaptation projects and assessing whether preservation value has been increased or not. Proper historical-stylistic evaluation requires temporal distance to observe societal and architectural trends, posing challenges for contemporary interventions.

A key challenge is the limited translation of ecological factors within the current LCA framework. Incorporating additional environmental indicators, such as biodiversity and planetary boundaries, could enhance the comprehensiveness of the assessment, either within the existing matrix or through a series of interconnected matrices. When adapting buildings for new uses, accepting higher environmental impacts may be necessary and justified if it secures the building's continued relevance and prevents demolition. The key is finding the right balance - making enough changes to ensure the building remains functional and relevant for contemporary needs while being mindful not to over-intervene.

REFERENCES

Agency for Castles and Culture. 2010. *Vejledning til vurdering af fredningsværdier*.

Andersen, R. 2023. *Informing Architectural Design Processes in a Circular Economy - the Quantification of Circular Construction*. Technical University of Denmark.

Bekendtgørelse Af Lov Om Bygningsfredning (Bygningsfredningsloven). 1986. LBK nr 81 af 10/02/1986

Bendsen, J. R., Morgen, M. A., & Lindhe, J. 2018. *Fredet: Bygningsfredning i Danmark 1918-2018* (1. edition). Strandberg Publishing.

Calvin, K. et al. 2023. *IPCC, 2023: Climate Change 2023: Synthesis Report*. IPCC, Geneva, Switzerland.

Cantacuzino, Sherban. 1975. *New Uses for Old Buildings*. New York, Whitney Library of Design.

Council of Europe. (n.d.). *Convention for the Protection of the Architectural Heritage of Europe (Granada, 1985)—Culture and Cultural Heritage*. Culture and Cultural Heritage.

Cramer, Johannes & Stefan Breitling. 2007. *Architecture in Existing Fabric - Planning, Design, Building*. Basel Switzerland; Birkhauser.

Dansk Arkitektur Center. 2011. *Arkitektur og Energirenovering - Det Murede Etagebyggeri fra 1920 til 1960, 2011*. Dansk Arkitektur Center.

Didriksen, K., & Tønnesen, A. 1997. *InterSAVE. Minestry of Environment and Energy and The National Forest and Nature Agency*.

Didriksen, K., Jensen, T., Høi, A., & Stenark, M. 2011. *SAVE – Kortlægning og registrering af bymiljøers og bygningers bevaringsværdi*. Ministry of Culture.

Douglas, James. 2006. *Building Adaptation – 2nd Edition*. Routledge.

ICOMOS. 1965. *The Venice Charter 1964 - International Charter for the Conservation and Restoration of Monuments and Sites*.

Ministry for Interior & Housing. 2021. *National Strategy for Sustainable Construction*.

Ministry for Social Affairs & Housing. 2024. *Ny aftale stiller ambitiøse klimakrav til nyt byggeri*.

Munch-Petersen, Pelle. 2020. *Circular Facade Design: The Tectonics of Circular Economy*. The Royal Danish Academy of Fine Arts, Schools of Architecture, Design and Conservation.

Munch-Petersen, Pelle, and Henriette Ejstrup. 2017. The Relations between Building Performance and Embedded Energy: Living and Sustainability. Eric An (ed.) *Living and Sustainability: An Environmental Critique of Design & Building Practices, Locally and Globally*, AMPS Proceedings Series 9: 66–74.

Planloven. 2018. Available at: https://www.retsinformation.dk/eli/lta/2020/1157

Plevoets, Bie & Koenraad Van Cleempoel. 2019. *Adaptive Reuse of the Built Heritage: Concepts and Cases of an Emerging Discipline*. Routledge & CRC Press.

Realdania. 2024. *Klimavisioner for modernismens bygningskultur*.

Realdania. 2024. *Renovering er bedst for klimaet - Konklusioner fra udviklingsprojektet Klimadata for renovering*.

Schjødt Worm et al. 2017. *Branchevejledning i LCA ved renovering*. Teknologisk Institut.

Stone, Sally. 2019. *UnDoing Buildings: Adaptive Reuse and Cultural Memory*. Milton, UNITED KINGDOM: Taylor & Francis Group.

Strike, James. 1994. *Architecture in Conservation: Managing Development at Historic Sites*. London: Routledge.

Stylsvig Madsen, Ulrik, Anne Beim, and Aske Reitz. 2015. *Værdiskabelse i Bygningsrenovering: "Bygningers Værdi Er Noget vi Skaber Sammen*. København: CINARK Kunstakademiets Arkitektskole.

United Nations. 2015. The Paris Agreement. UNFCCC